# A.I.M.S. (AUTOMATED INTERVENTION AND MASTERY SYSTEM): A WEB-BASED EDUCATIONAL PLATFORM WITH AI-DRIVEN ASSESSMENT AND DYNAMIC REMEDIATION

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# Chapter 1

# INTRODUCTION

## Background of the Study

Technology is becoming a part of education systems across the globe to aid in teaching, learning and evaluation. Educational technologies that are powered by Artificial Intelligence (AI) have demonstrated the potential to deliver personalized learning experiences, adaptive feedback, and automated assessment capabilities capable of assisting educators to respond to the needs of different learners and enhance their academic performance. Studies have shown that AI-based tools have the potential to make instruction more efficient, customize learning strategies to accommodate personalized learning processes, and facilitate informed decision-making in academic environments (Merino-Campos, 2025; Marcos, 2026).

The use of technology in basic education in the Philippine context has been increasing, and the COVID-19 pandemic has contributed to the adoption of flexible learning modalities. Nevertheless, age-old challenges of digital divide, uneven access to learning resources, poor infrastructure, and manual assistance with evaluation and module distribution continue to be a problem of many public schools (Bustillo & Aguilos, 2022; Balase & Paglinawan, 2025). Teachers tend to use non-standardization means like group discussions to communicate modules and have to score and mark student performance manually, which deprives them of time to teach and provide individualized intervention. In addition, research on educational technology in the Philippines also states that although

digital tools such as LMS are employed, the lack of connectivity and digital skills hinders their maximum functionality (Colegado, 2025).

Those difficulties leave the necessity to develop more intelligent and systematic solutions that can be used to centralize content delivery, automate assessment procedures, and facilitate personalized intervention of learners. This requirement is especially appropriate to AI-based systems, which include automated question generation, dynamically-adaptive assessments, and analytics dashboards that can inform teaching and correction. The positive effect of such systems on engagement and learning outcomes is supported in the international literature in case such systems are implemented thoughtfully.

That being the case, the given study presents A.I.M.S. (Automated Intervention and Mastery System) an AI-powered web-based educational solution aimed at supporting the work of the public school teachers simplifying the work with modules and offering automated evaluation and dynamic remediation based on the performance of individual students. A.I.M.S. is expected to help decrease the administrative load, enhance the correctness of the tracking of performance, and assist in making timely decisions in instructions, by incorporating the use of features like Module Management, Automatic assessments, AI-powered quiz generation, automated grading, competency-based progression, and performance tracking into it. It is projected that this implementation of the A.I.M.S. platform will fill the recorded gaps in the use of technologies, increase the workflow of the teachers, and become stronger in terms of the results of the learners in governmental schools due to intelligent automation and the data-driven instructions.

**Objective of the Study**

The general objective of the study aims to design and develop a web-based application titled "A.I.M.S. (Automated Intervention and Mastery System): A Web-Based Educational Platform with AI-Driven Assessment and Dynamic Remediation".

Specifically, this aims the following:

1. design and develop a web-based application with the following technical features:
    1.1. centralized digital library for module distribution and management.
    1.2. ai-powered quiz generator with instructor validation controls.
    1.3. automated grading module with customizable rubrics.
    1.4. mastery-lock system for automated student remediation.
    1.5. student performance dashboard for real-time academic tracking.
2. test the functionality of the features mentioned above
3. evaluate the system using ISO 25010 standardize evaluation tool in terms of
    3.1. Functionality Suitability
    3.2. Performance Efficiency
    3.3. Usability
    3.4. Compatibility
    3.5. Reliability
    3.6. Security
    3.7. Maintainability
    3.8. Portability

**Significance of the Study**

A.I.M.S. (Automated Intervention and Mastery System): A Web-Based Educational Service with AI-Based Assessment and Dynamic Remediation is marketed to be of substantial and positive importance to the following groupings:

**Teachers**. Such system will help the teachers to have better control of learning modules, automation of assessment, production of adaptive questions and real time tracking of student performance. This will allow teachers to concentrate more on instruction delivery, individual support and intervention on students who require further guidance by lessening the administrative burdens, as well as, offering timely analytics.

**Students**. A.I.M.S. let students study learning modules and tests on the internet and get immediate feedback, as well as enjoy AI-based dynamic correction based on their personal performance. This promotes self-directed learning, enables learners to gain awareness of their areas of weakness and areas of strength and mastery of learning content.

**Researchers**. The research offers empirical evidence concerning the use of the AI-based web-based learning systems in the Philippine public schools environment. The effectiveness of the system to enhance the learning results, improve the process of assessments, and assist personalized learning can be studied by the researchers as the part of the overall literature on educational technology and AI-assisted learning.

**Future Researchers**. The results and development experience of A.I.M.S. can be used as a base in the future studies of intelligent learning platforms. Researchers may extend on the structure of the system in future, add features or investigate other AI

applications in education, creating new innovations to improve the teaching and learning experience in different teaching and learning environments.

**Scope and Limitation of the Study**

The paper is devoted to the design, development and testing of A.I.M.S. (Automated Intervention and Mastery System): A Web-Based Educational Platform with AI-Driven Assessment and Dynamic Remediation that is supposed to be used in the local school environment. Some of its core features covered by the system are module management which gives teachers the ability to post and distribute learning materials on-line or in printed form, automated evaluation which provides recording and scoring of student responses, automatic generation of assessment questions but with the ability of the teacher to control them, module distribution in a standardized form so that official learning material is made available to all students and automated grading with customizable rubrics to provide feedback and custom scoring. The system will be tested according to the ISO 25010 standard that determines functionality, performance, usability, compatibility, reliability, security, maintainability, and portability.

The study is limited in some ways in spite of its elaborate design. A.I.M.S. is mainly designed to serve the population schools and might have to be modified to suit the use in the privates or tertiary institutions. The online services of it require the stable connection of the internet, and the offline services are restricted with printing modules. The paper does not consider the long-term impacts on the academic performance of students because the analysis is on the system usability, functionality and short-term effectiveness. Also, the

automatic question generator is based on pre-operative algorithms, and it might be necessary to involve the teacher in order to meet the assessment requirements that are specific and contextual. These limitations stipulate the limits of the research as well as present a clear guideline towards assessing the functionality and effectiveness of the system in the target environment.

**Definition of Terms**

For a clearer understanding of the study, the following terms are defined both contextually based on existing literature and operationally as they are used in the context of A.I.M.S.

**AI-Powered Quiz Generator.** Contextually, this refers to the use of neural question generation techniques and natural language processing to automatically create educationally valuable assessment items from learning materials, ensuring both fluency and pedagogical relevance (Bulathwela et al., 2023).

Operationally, in this study, it refers to the module in A.I.M.S. that uses Google Gemini API to automatically generate multiple-choice, true/false, and short-answer quizzes aligned with DepEd competencies from teacher-uploaded documents.

**Automated Grading Module.** Contextually, it refers to the use of algorithms and natural language processing to evaluate student responses and assign scores based on predefined assessment criteria, reducing manual grading workload while maintaining consistency (Ramesh & Sanampudi, 2022).

Operationally, it is the system component in A.I.M.S. that evaluates quiz responses according to teacher-defined customizable rubrics, providing immediate automated score computation and instant feedback to students.

**Centralized Digital Library.** Contextually, this is an integrated online platform within a learning management system that organizes and distributes educational materials, enabling structured access to course content and resources (Bradley, 2021).

Operationally, it is the repository feature in A.I.M.S. that allows teachers to upload and organize learning modules by subject and grade level, enabling students to securely access, search, and download educational materials.

**Mastery-Lock System.** Contextually, in competency-based education, this is a progression mechanism where students acquire knowledge and complete formative assessments, proceeding to new content only after demonstrating mastery of prerequisite material through corrective pathways and retesting (Persky & Hughes, 2022).

Operationally, it is the conditional progression feature in A.I.M.S. that restricts access to subsequent lessons until students achieve 70 percent or higher on previous assessments, triggering automated remediation interventions for non-mastery.

**Student Performance Dashboard.** Contextually, it is a visual analytics interface that provides actionable insights into learning behaviors and academic progress through data-driven visualizations, enabling students to regulate their learning activities effectively (Susnjak et al., 2022).

Operationally, it is the analytics component in A.I.M.S. that displays interactive charts tracking student engagement metrics, quiz completion rates, mastery scores, and progression status for real-time academic monitoring.

**ISO 25010.** Contextually, it is the international software quality standard that defines eight characteristics for evaluating system and software products: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability (ISO/IEC, 2011).

Operationally, it is the standardized evaluation framework used by the researchers to rigorously assess A.I.M.S.' system quality, effectiveness, and user satisfaction through surveys and functional testing during the validation phase.

# Chapter 2

## REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the review of related literature and studies relevant to the development of A.I.M.S. (Automated Intervention and Mastery System). The discussion encompasses both foreign and local literature organized thematically around the five technical features of the proposed system: centralized digital library for module distribution and management, AI-powered quiz generator with instructor validation controls, automated grading module with customizable rubrics, mastery-lock system for automated student remediation, and student performance dashboard for real-time academic tracking.

### Web-Based Centralized Digital Library for Module Distribution and Management

The centralized digital library has emerged as a foundational component of modern learning management systems (LMS). According to Bradley (2021), LMS platforms that incorporate organized digital repositories significantly improve accessibility to course materials, enable structured content delivery, and reduce the administrative burden associated with manual material distribution. Bradley's study, which analyzed LMS use in online instruction across multiple institutional contexts, found that educators who deployed centralized module repositories reported improved learner engagement and more efficient course management compared to traditional paper-based methods, establishing digital repository integration as a key determinant of instructional effectiveness in web-based learning environments.

In the Philippine context, the challenge of uneven access to learning materials and inadequate infrastructure has long impeded effective teaching in public schools. Bustillo and Aguilos (2022) documented persistent barriers to educational technology adoption, noting that teachers are often compelled to distribute printed modules manually—an approach that is both inefficient and prone to inconsistencies due to limited reproduction resources and geographical accessibility constraints. Their study highlights that digital platforms capable of centralizing and streamlining module distribution offer a practical and scalable solution to the longstanding inequities in educational resource access across Philippine public schools, making a compelling case for LMS-integrated digital libraries as an institutional priority.

Web-based module management systems address these challenges by enabling teachers to upload, categorize, and update learning materials electronically, while students access content through a standardized platform at any time and from any location (Bradley, 2021). Such systems support detailed usage tracking, allowing educators to monitor student access patterns and identify learners who have not engaged with assigned materials—a feature especially valuable for early identification of at-risk students. Taken together, the evidence from Bradley (2021) and Bustillo and Aguilos (2022) affirms that a centralized digital library within A.I.M.S. is both pedagogically justified and contextually responsive to the documented access challenges in Philippine educational settings.

**AI-Powered Quiz Generator with Instructor Validation Controls**

Artificial intelligence has fundamentally transformed how educational assessments are designed and administered, particularly through automated question generation. Bulathwela et al. (2023) conducted an extensive review of neural question generation (NQG) systems for educational purposes, evaluating rule-based and neural architectures across multiple studies. Their analysis demonstrated that transformer-based NQG models are capable of producing linguistically fluent and pedagogically appropriate questions from source texts, generating diverse item formats—including multiple-choice, true/false, and short-answer—with quality comparable to human-authored questions when guided by sufficient contextual parameters. This capability positions AI-powered generators as viable tools for reducing teacher workload in formative assessment design.

Recent research has further validated the use of large language models (LLMs) in automated question generation for educational domains. Alamoudi et al. (2025) compared three methodological approaches—template-based structured ontology generation, LLM-based structured ontology generation, and LLM-based flat concept list generation—using BERT Precision, Recall, F1-score, and semantic similarity as performance metrics. Their findings indicate that the LLM-based flat concept list approach achieved the highest semantic similarity (0.567) and the most balanced precision-recall performance. Significantly, the authors noted that unstructured LLM generation occasionally produced hallucinated or topically unrelated questions, reinforcing the necessity of instructor validation controls in any AI-powered quiz generation system to ensure accuracy, curriculum alignment, and pedagogical relevance before deployment.

In the Philippine educational landscape, the administrative challenges of assessment preparation compound existing systemic difficulties. Balase and Paglinawan (2025) identified that teachers in Philippine public schools face substantial workload burdens in preparing, distributing, and scoring assessments manually, particularly in large classes where formative evaluation is frequent. This workload significantly limits teachers' capacity to focus on instructional quality and individualized student support. The integration of an AI-powered quiz generator with instructor validation controls in A.I.M.S. directly addresses this challenge by automating item generation while preserving teacher oversight—ensuring both efficiency and educational integrity throughout the assessment pipeline.

**Automated Grading Module with Customizable Rubrics**

Automated grading systems have gained considerable traction in educational technology research, with evidence demonstrating their effectiveness in evaluating student-authored responses with consistency, speed, and scalability. Ramesh and Sanampudi (2022) conducted a comprehensive systematic literature review of automated essay scoring (AES) systems, analyzing over one hundred studies to evaluate the accuracy, scalability, and applicability of machine learning and natural language processing approaches to written response grading. Their review found that deep learning models—particularly transformer architectures such as BERT—achieved high agreement with human raters on holistic scoring, and that AES systems substantially reduced grading time while

maintaining measurement validity, making them practical for large-scale educational deployment.

Beyond holistic scoring, the incorporation of customizable rubrics into automated grading systems is critical for aligning AI-generated scores with instructor-defined pedagogical standards. García-Varela et al. (2025) investigated the reliability and fairness of ChatGPT as an automated essay scoring tool, demonstrating that when provided with explicit, normalized rubrics and standardized output formats at low temperature settings, the model achieved perfect grading consistency across ten independent evaluation sessions. Their study also developed a scalable algorithm that automatically generates normalized grading rubrics and decision tables from any open-ended question with minimal instructor input. These findings confirm that the combination of AI grading capability with customizable rubric structures yields consistent, transparent, and educationally sound assessment outcomes.

In Philippine educational settings, where teachers routinely manage large classes, the time consumed by manual evaluation substantially limits the instructional time available for lesson preparation and individualized student support. Colegado (2025), in a scoping review of digital innovations across Philippine K–12 science education, found that while learning management systems such as Google Classroom and Quipper enabled structured content delivery and assessment monitoring, their consistent use was largely confined to urban and better-resourced schools. In underserved and rural settings, inadequate infrastructure and limited teacher preparation continued to constrain the effectiveness of digital assessment tools, leaving many educators without scalable means

to evaluate student work systematically. These findings highlight a persistent gap in Philippine educational technology adoption and directly motivate the inclusion of an automated grading module with customizable rubrics in A.I.M.S., which enables teachers to administer and evaluate assessments at scale while maintaining instructional quality through teacher-defined, adaptable scoring criteria—regardless of school resource level.

**Adaptive Mastery-Lock System for Automated Student Remediation**

Mastery-based learning is a pedagogical model grounded in the principle that all students can achieve academic proficiency when given sufficient time, appropriate instructional support, and the opportunity to demonstrate competency before progressing to subsequent content. Persky and Hughes (2022) conducted a comprehensive review of mastery learning principles and their practical application in varied educational contexts, finding that mastery-based instruction consistently outperformed traditional time-based delivery models in promoting long-term content retention, reducing achievement gaps, and improving student confidence. Their review emphasized that effective mastery learning frameworks require clearly defined performance thresholds, systematic corrective feedback mechanisms, and structured opportunities for retesting—components that are directly embedded in the mastery-lock system proposed within A.I.M.S.

The adaptive dimension of the mastery-lock system distinguishes it from static competency gates by dynamically adjusting remediation content based on individual student performance data. Rather than merely blocking progression at a fixed score

threshold, adaptive mastery-lock systems analyze patterns in student errors and misconceptions to deliver targeted remediation activities aligned with specific learning gaps. This integration of adaptive remediation with competency-based progression has been validated in the broader learning analytics literature, where data-driven feedback mechanisms have been shown to improve students' self-regulatory behavior and academic performance (Susnjak et al., 2022). By connecting performance data to automated intervention triggers, the A.I.M.S. mastery-lock mechanism ensures that progression decisions are both consistent and educationally purposeful.

In the Philippine educational context, class size constraints and limited instructional support time make individualized intervention challenging to sustain manually (Balase & Paglinawan, 2025). Balase and Paglinawan's study on educational technology adoption barriers in Philippine public schools found that while adaptive learning tools hold significant institutional promise, their effective implementation requires structural integration with automated performance tracking and decision-support systems that do not depend on continuous teacher mediation. The A.I.M.S. mastery-lock system addresses this implementation gap by automating progression decisions and corresponding remediation delivery, reducing dependency on teacher-initiated intervention while maintaining pedagogical consistency and competency-based rigor across diverse classroom environments.

Empirical evidence from the Philippine basic education context further affirms the value of structured remediation in supporting learner progression. Jumao-as et al. (2025)

investigated the effectiveness of a reading remediation program implemented in a DepEd elementary school in Ozamiz City, Misamis Occidental, finding that Grade 4 pupils identified as having frustrated reading levels through the Philippine Informal Reading Inventory (PHIL-IRI) advanced to the instructional reading level following the systematic remediation program. The study demonstrated a statistically significant improvement in reading comprehension, validating that targeted, competency-specific intervention embedded within a structured instructional cycle produces measurable learning gains. These findings from the Philippine basic education context provide direct empirical support for the mastery-lock design principle in A.I.M.S., confirming that systematic, criterion-referenced remediation—when consistently applied—effectively closes performance gaps and moves learners toward academic proficiency.

**Web-Based Student Performance Dashboard for Real-Time Academic Tracking**

Learning analytics dashboards have emerged as powerful tools for converting raw educational data into actionable insights that support both student self-regulation and teacher instructional decision-making. Susnjak et al. (2022) investigated how learning analytics dashboards can provide actionable insights to learners by synthesizing engagement metrics, quiz performance data, and learning behavior logs into interpretable visualizations. Their study found that dashboards displaying timely and contextually interpretable performance information significantly improved students' awareness of their academic trajectories, encouraged proactive study behaviors, and enabled educators to identify and support at-risk learners before performance deteriorated critically—

positioning the performance dashboard as an indispensable component of any AI-enhanced educational platform.

The design and usability of learning analytics dashboards substantially influence their adoption rate and educational efficacy in real classroom settings. De Vreugd et al. (2024) conducted a sequential mixed-methods study involving needs assessments, think-aloud usability interviews, and perceived usefulness evaluations to design a dashboard supporting student self-regulation of study behavior in higher education. Their findings revealed that students and instructors agreed on the relevance of core performance constructs—including quiz completion rates, engagement levels, and module access frequency—but differed in how they prioritized these data dimensions. The study also demonstrated that providing interpretive reference frames alongside raw data significantly enhanced the dashboard's perceived usefulness, offering critical design guidance for the A.I.M.S. student performance dashboard.

A systematic review by Kaliisa et al. (2024), synthesizing findings from 38 research studies on learning analytics dashboards, found that these tools consistently and significantly improved student participation and engagement, with several studies reporting medium to large effect sizes. However, the review also noted that evidence for direct impact on academic achievement remains limited and methodologically inconsistent, largely due to small sample sizes, self-selection biases, and the conflation of dashboard access with meaningful use. In the Philippine educational context, where data-driven instruction remains nascent and manual performance tracking is still common practice, the

introduction of a web-based student performance dashboard through A.I.M.S. represents a substantive step toward evidence-informed teaching and real-time academic monitoring at the classroom level.

In the Philippine educational context, where data-driven instruction and digital performance monitoring remain emergent practices, locally developed system frameworks have begun to demonstrate the feasibility of web-based academic tracking tools in public school settings. Gajardo and Balahadia (2025) developed the SMART eLearning framework—a conceptual web portal for senior high school students in the Philippines designed to integrate diagnostic and achievement assessments with a backtracking algorithm that systematically traces incorrect answers to underlying knowledge gaps. The framework incorporates automated quizzes, performance dashboards, and real-time progress tracking to deliver immediate feedback to both learners and educators, directly addressing persistent underperformance in core subjects as reflected in PISA and the National Achievement Test (NAT). This locally situated work validates the demand for integrated performance dashboard functionality within Philippine school systems and affirms the design rationale for the student performance dashboard component of A.I.M.S.

**Synthesis of the Reviewed Literature**

The reviewed literature collectively demonstrates that each of the five core features of A.I.M.S. is grounded in a robust and growing body of empirical evidence. Centralized digital libraries address persistent access and distribution inequities in public school

environments (Bradley, 2021; Bustillo & Aguilos, 2022), while AI-powered quiz generators substantially reduce the cognitive and administrative burden of assessment creation (Bulathwela et al., 2023; Alamoudi et al., 2025). Automated grading systems promote evaluation consistency and scalability through machine learning and rubric-guided assessment (Ramesh & Sanampudi, 2022; García-Varela et al., 2025), and mastery-lock mechanisms provide structured, competency-based progression proven to reduce learning gaps (Persky & Hughes, 2022; Jumao-as et al., 2025). Performance dashboards complete the instructional feedback loop by delivering real-time visibility into student progress (Susnjak et al., 2022; de Vreugd et al., 2024; Kaliisa et al., 2024; Gajardo & Balahadia, 2025).

A consistent theme across both foreign and local literature is that the greatest educational impact is achieved when multiple intelligent features are integrated cohesively within a single platform, rather than deployed as fragmented standalone tools that create workflow disruptions and adoption resistance (Balase & Paglinawan, 2025; Colegado, 2025). The reviewed studies demonstrate that while individual features—quiz generation, automated grading, mastery progression—have been validated in isolation, no existing platform fully integrates the complete instructional and assessment cycle in the manner envisioned in A.I.M.S. This integration gap represents the primary research and design opportunity that the present study addresses.

Overall, the literature reviewed in this chapter provides strong theoretical, empirical, and contextual justification for the design and development of A.I.M.S. as a

unified, AI-driven educational platform tailored to the Philippine public school context. The evidence base confirms that the proposed features are not merely technically feasible but also pedagogically sound, contextually responsive, and aligned with contemporary best practices in educational technology research. By operationalizing verified AI capabilities within a coherent, teacher-centered system architecture, A.I.M.S. represents a research-supported response to documented gaps in Philippine digital learning environments—integrating access management, automated assessment, competency-based progression, and performance analytics into a single, scalable educational system.

**Table 1**

**Matrix of Related Literature and Studies**

| **RRL** | **Features** | | | | |
|---|---|---|---|---|---|
| **Studies/System** | **Centralized Digital Library** | **AI-Powered Quiz Generator** | **Automated Grading Module** | **Mastery-Lock System** | **Student Performance Dashboard** |
| Optimizing Automated Question Generation for Educational Assessments – Alamoudi et al. (2025) | ✗ | ✓ | ✓ | ✗ | ✗ |
| Learning Management System (LMS) Use with Online Instruction – Bradley (2021) | ✓ | ✗ | ✗ | ✗ | ✗ |
| Review on Neural Question Generation for Education Purposes – Bulathwela et al. (2023) | ✓ | ✓ | ✗ | ✗ | ✗ |
| Learning Analytics Dashboard Design and Evaluation to Support Student Self-Regulation – de Vreugd et al. (2024) | ✗ | ✗ | ✗ | ✗ | ✓ |
| ChatGPT as a Stable and Fair Tool for Automated Essay Scoring – García-Varela et al. (2025) | ✗ | ✗ | ✓ | ✗ | ✗ |

| | | | | | |
|---|---|---|---|---|---|
| Have Learning Analytics Dashboards Lived Up to the Hype? A Systematic Review – Kaliisa et al. (2024) | ✗ | ✗ | ✗ | ✗ | ✓ |
| A Practical Review of Mastery Learning – Persky & Hughes (2022) | ✗ | ✓ | ✓ | ✗ | ✗ |
| An Automated Essay Scoring Systems: A Systematic Literature Review – Ramesh & Sanampudi (2022) | ✗ | ✗ | ✓ | ✗ | ✗ |
| Learning Analytics Dashboard: A Tool for Providing Actionable Insights to Learners – Susnjak et al. (2022) | ✗ | ✗ | ✗ | ✓ | ✓ |
| The Challenges of Modular Learning in the Wake of COVID-19: A Digital Divide in the Philippine Countryside Revealed – Bustillo & Aguilos (2022) | ✓ | ✗ | ✗ | ✗ | ✗ |
| Voices from the Field: Teachers' Narratives and Reflections on Integrating the DepEd LMS – Balase & Paglinawan (2025) | ✓ | ✗ | ✗ | ✓ | ✗ |
| Digital Innovations in Science Education in the Philippines: A Scoping Review – Colegado (2025) | ✓ | ✗ | ✗ | ✗ | ✗ |
| Smart eLearning: A Framework Development of a Web Portal for Data-Driven Assessment – Gajardo & Balahadia (2025) | ✗ | ✗ | ✗ | ✗ | ✓ |
| Reading Remediation Program and Grade 4 Pupils' Reading Comprehension – Jumao-as et al. (2025) | ✗ | ✗ | ✗ | ✓ | ✗ |

**Synthesis**

The matrix of related literature and studies above presents a comparative overview of existing systems and studies in relation to the five core technical features of the proposed A.I.M.S. platform. The studies mapped in the matrix span both foreign and local literature, covering digital library systems and LMS integration (Bradley, 2021; Bustillo & Aguilos, 2022), AI-driven and LLM-based quiz generation methods (Bulathwela et al., 2023; Alamoudi et al., 2025), automated grading and AES approaches with rubric customization (Ramesh & Sanampudi, 2022; García-Varela et al., 2025), mastery-based learning and competency progression frameworks (Persky & Hughes, 2022; Jumao-as et al., 2025), and

learning analytics dashboard design and evaluation (Susnjak et al., 2022; de Vreugd et al., 2024; Kaliisa et al., 2024; Gajardo & Balahadia, 2025).

Several studies demonstrate the value of centralized digital libraries for efficient and equitable module distribution, while others validate AI-powered assessment tools for reducing teacher workload and improving evaluation consistency. Competency-based progression supported by mastery learning principles is affirmed as an effective instructional strategy across diverse learner populations, and learning analytics dashboards are consistently associated with improved student participation and engagement. Local Philippine studies document systemic barriers to technology adoption and underscore the demand for integrated, automated digital solutions in public school settings (Bustillo & Aguilos, 2022; Balase & Paglinawan, 2025; Colegado, 2025; Jumao-as et al., 2025; Gajardo & Balahadia, 2025).

Overall, the synthesis reveals that while the individual features addressed by A.I.M.S. are present in existing systems and have been independently validated in the literature, no single platform currently integrates all five functionalities—centralized digital library, AI quiz generation, automated grading, mastery-lock progression, and performance dashboard—into a unified, AI-driven educational environment. This integration gap constitutes the primary research contribution of A.I.M.S. and affirms the novelty and practical significance of the proposed system as a response to documented needs in the Philippine educational technology landscape.

## REFERENCES

Alamoudi, S., Al Khuzayem, L. A., & Jamal, A. (2025). Optimizing automated question generation for educational assessments: A semantic analysis of LLMs with structured and unstructured ontologies. *Engineering, Technology & Applied Science Research, 15*(3), 23664–23671. https://doi.org/10.48084/etasr.10662

Balase, M. M. H., & Paglinawan, J. L. (2025). Voices from the field: Teachers' narratives and reflections on integrating the DepEd Learning Management System (LMS) in teaching and professional development. *International Journal of Scientific and Management Research, 8*(10), 147–159. https://doi.org/10.37502/ijsmr.2025.81011

Bradley, V. M. (2021). Learning management system (LMS) use with online instruction. *International Journal of Technology in Education, 4*(1), 68–92. https://doi.org/10.46328/ijte.36

Bulathwela, S., Pérez-Ortiz, M., Holloway, C., Cukurova, M., & Shawe-Taylor, J. (2023). Review on neural question generation for education purposes. *International Journal of Artificial Intelligence in Education, 33*(4), 691–717. https://doi.org/10.1007/s40593-023-00374-x

Bustillo, E., & Aguilos, M. (2022). The challenges of modular learning in the wake of COVID-19: A digital divide in the Philippine countryside revealed. *Education Sciences, 12*(7), 449. https://doi.org/10.3390/educsci12070449

Colegado, J. C. (2025). Digital innovations in science education in the Philippines: A scoping review of teaching practices and tools. *International Journal of Research*

*and Innovation in Social Science, 9*(3S), 6549–6556. https://doi.org/10.47772/IJRISS.2025.903SEDU0479

de Vreugd, L., van Leeuwen, A., Jansen, R., & van der Schaaf, M. (2024). Learning analytics dashboard design and evaluation to support student self-regulation of study behaviour. *Journal of Learning Analytics, 11*(3), 249–262. https://doi.org/10.18608/jla.2024.8529

Gajardo, M. R., & Balahadia, F. F. (2025). Smart eLearning: A framework development of a web portal for data-driven assessment and module recommendation for senior high school students using backtracking algorithm. *International Journal of Science, Technology, Engineering and Mathematics, 5*(3), 22–39. https://doi.org/10.53378/ijstem.353233

García-Varela, F., Nussbaum, M., Mendoza, M., Martínez-Troncoso, C., & Bekerman, Z. (2025). ChatGPT as a stable and fair tool for automated essay scoring. *Education Sciences, 15*(8), Article 946. https://doi.org/10.3390/educsci15080946

International Organization for Standardization. (2011). *Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—System and software quality models* (ISO/IEC Standard No. 25010:2011). https://www.iso.org/standard/35733.html

Jumao-as, J., Fuentes, J. L., Lacar, L., Fuertes, G. J., Erag, C., & Tagaylo, C. (2025). Reading remediation program and Grade 4 pupils' reading comprehension. *Asian Journal of Education and Social Studies, 51*(6), 365–377. https://doi.org/10.9734/ajess/2025/v51i62000

Kaliisa, R., Misiejuk, K., López-Pernas, S., Khalil, M., & Saqr, M. (2024). Have learning analytics dashboards lived up to the hype? A systematic review of impact on students' achievement, motivation, participation and attitude. In *Proceedings of the 14th Learning Analytics and Knowledge Conference (LAK '24)* (pp. 295–304). ACM. https://doi.org/10.1145/3636555.3636884

Marcos, L. T. (2026). A systematic review on artificial intelligence in education: Opportunities, challenges, and ethical implications. *Preprints*. https://doi.org/10.20944/preprints202601.0448.v1

Merino-Campos, C. (2025). The impact of artificial intelligence on personalized learning in higher education: A systematic review. *Trends in Higher Education, 4*(2), Article 17. https://doi.org/10.3390/higheredu4020017

Persky, A. M., & Hughes, M. L. (2022). A practical review of mastery learning. *American Journal of Pharmaceutical Education, 86*(10), Article 8906. https://doi.org/10.5688/ajpe8906

Ramesh, D., & Sanampudi, S. K. (2022). An automated essay scoring systems: A systematic literature review. *Artificial Intelligence Review, 55*(3), 2495–2527. https://doi.org/10.1007/s10462-021-10068-2

Susnjak, T., Ramaswami, G. S., & Mathrani, A. (2022). Learning analytics dashboard: A tool for providing actionable insights to learners. *International Journal of Educational Technology in Higher Education, 19*(1), Article 12. https://doi.org/10.1186/s41239-021-00313-7